Cell–cell interactions in *Dictyostelium* development

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The development of an organism requires extensive cell–cell communication; however, little is known about the signals transmitted among differentiating cells. Observations of Dictyostelium amoebae reveal that transmembrane signaling systems have been highly conserved in evolution. The signals that cause these cells to differentiate are processed by mechanisms similar to those that process sensory and hormonal stimuli in higher animals.

Recent studies of the social amoeba *Dictyostelium* are contributing to our understanding of the role of cell–cell interactions in development. When deprived of nutrients, these cells cease growth and initiate a program in which 10^5 single amoebae form a multicellular organism. The aggregation of the cells is organized by extracellular adenosine 3', 5' cyclic phosphate (cAMP) which mediates chemotaxis and oscillatory cell–cell signaling (Fig. 1). The signaling derives from the capacity of each cell to synthesize and secrete cAMP in response to cAMP stimulation. Within aggregation centers, cAMP levels spontaneously oscillate; each peak initiates a wave of cAMP that propagates through the cell monolayer. The leading edge of each passing cAMP wave provides a gradient that briefly orients the chemotactically sensitive cells towards the aggregation center. Cells move up the gradient for several minutes until the peak of the wave reaches the position of the cell. The cells then move randomly until the next wave elicits another coordinated movement step. Within a few hours, these movements result in the formation of a multicellular structure. During the following day, it undergoes a series of morphogenetic changes and a pattern forms along its axis. The cells differentiate into the two cell types, vacuolated nonviable stalks and durable spores, which constitute the mature fruiting body[1,2].

Figure 2 is a schematic diagram of the cell–cell signaling system. High-affinity surface cAMP receptors mediate the chemotactic response as well as activation of adenylate cyclase. Secreted cAMP diffuses to nearby cells where it activates the same responses. It also binds to the secreting cell, creating a positive feedback loop. Following an initial discharge lasting several minutes, uncontrolled production of cAMP is prevented by an adaptation mechanism that reversibly attenuates synthesis. An extracellular phosphodiesterase then lowers cAMP towards basal levels. This combination of opposing molecular forces makes the system excitable. This excitability accounts for the spontaneous cAMP oscillations that occur at the aggregation centers and also in synchronized suspensions of cells[3].

It has become clear that this cell–cell signaling system plays a central role in development, acting as a timer and controlling morphogenesis and pattern formation[4–9]. Cyclic AMP, together with morphogens such as DIF (differentiation-inducing factor), regulates the classes of genes expressed at each stage of development and ensures that cells differentiate into prestalk and prespore cells in a pattern along the axis of the structure. This article will first describe the components of the signaling system, then discuss its role in gene expression, morphogenesis and pattern formation.

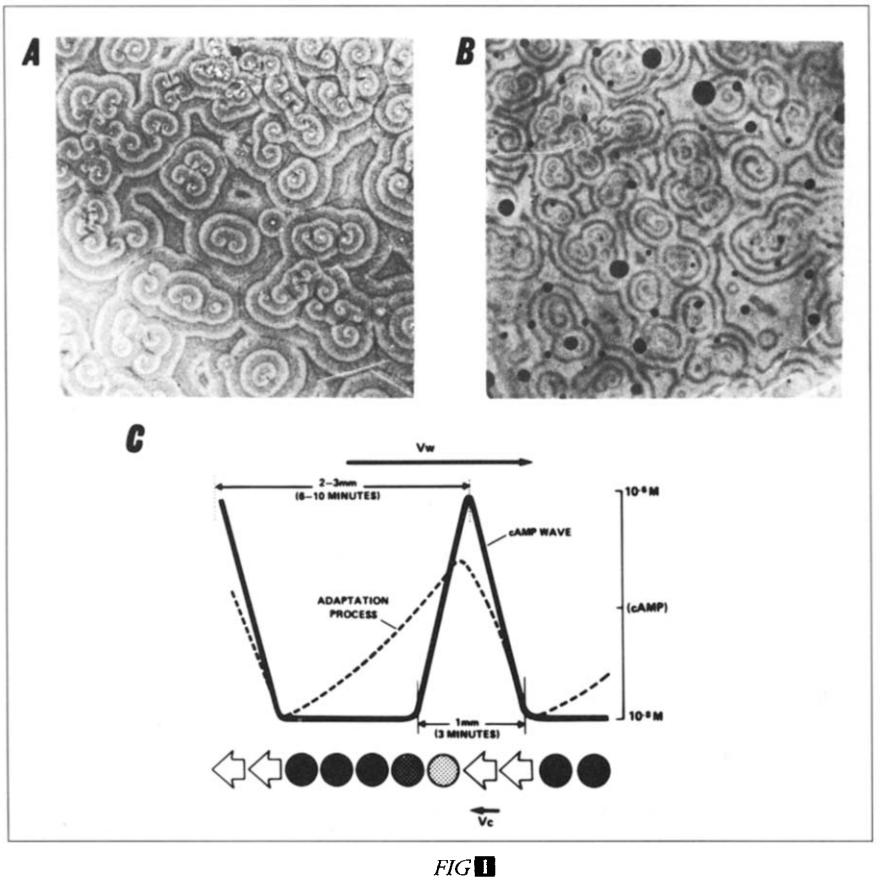

FIG **1**

Chemotaxis and cAMP signaling mediate aggregation. (A) Dark-field photograph of aggregating cells at the 5 hour stage of development. Territories containing 1 million cells are actually 1–2 cm in diameter. (B) Fluorographic image of cAMP waves within a monolayer of aggregating cells. (C) Dynamics of signal relay and chemotaxis. The heavy line represents the cAMP concentration. Symbols represent a single radial line of cells: open arrows, cells moving toward the center; shaded circles, randomly oriented cells; arrow vectors, the speed and direction of motion of the cAMP wave (V_w = 300 μm min^{-1}) and the moving cells (V_c = 20 μm min^{-1}).

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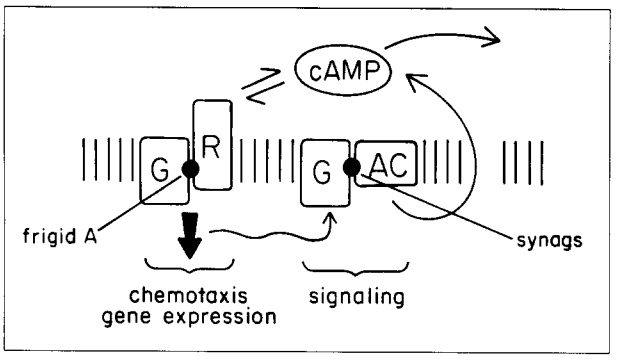

FIG 2

Diagram of signal transduction pathways. Shown are interactions among receptor (R), guanine nucleotide-binding protein(s) (G), and adenylate cyclase (AC); *frigid A* mutations block all signal transduction events as shown; *synag* mutations specifically prevent cAMP signaling as shown.

G-protein-linked signal transduction pathways

Multiple G proteins

The coupling of surface receptors to G proteins, a conserved family of guanine nucleotide-binding proteins, has emerged as a major signaling mechanism involved in sensory and hormone responsiveness. Perhaps the most significant lesson we are learning from studies of *Dictyostelium* is that G-protein-linked pathways are also involved in the regulation of gene expression via cell–cell interactions. The G proteins consist of heterotrimeric complexes that dissociate upon activation. The α-subunit binds and hydrolyses GTP; the βγ-subunits are required for recognition of the complex by the occupied receptor[10]. Two homologues of the α-subunits, designated $G_{\alpha 1}$ and $G_{\alpha 2}$, which are 45% identical with mammalian G proteins, have been identified[11], and one homologue of the β-subunit that is 63% identical to human β has also been found[12]. Several standard assays of G-protein function have been applied in this system. First, in isolated membranes, the affinity of receptors for cAMP is lowered by GTP and GDP[13]. Second, adenylate cyclase is stimulated by GTP analogues[14]. Mutations can independently affect each of these functions, suggesting that there are multiple classes of these signal-transducing proteins (Fig. 2): *frigid A* mutations interfere with the regulation of cAMP-binding affinity; *synag* mutations prevent GTP stimulation of adenylate cyclase[14–16].

These mutations are a subset of the approximately 100 complementation groups that display aggregation-minus phenotypes: *frigid A* mutants have no detectable chemotactic or signaling responses and cannot be induced to differentiate; *synag* mutants have specifically lost the signaling response but can carry out chemotaxis and can differentiate if supplied with cAMP. Further studies of *frigid A* alleles showed that the $G_{\alpha 2}$ gene carries deletions or point mutations (Ref. 17 and G. Pitt *et al.*, in preparation) and that a wild-type phenotype can be restored to these mutants by transformation with a wild-type $G_{\alpha 2}$ gene (A. Kumagai *et al.*, in preparation). Thus, $G_{\alpha 2}$ is a key component of a major signal transduction pathway. As yet, the molecular defect in a *synag* mutant has not been defined. A soluble protein from wild-type cells that can restore GTP stimulation to the adenylate cyclase in the *synag* 7 mutant is currently being purified[14].

Surface cAMP receptors

Figure 3A is a working model for the structure of the surface cAMP receptor[18]. It contains seven putative membrane-spanning domains, the amino terminus is extracellular, and a serine-rich carboxy-terminal domain is intracellular. There are prolines and charges within the membrane-spanning domains that may contribute to cAMP binding. All these features are common to the class of G-protein-linked receptors that

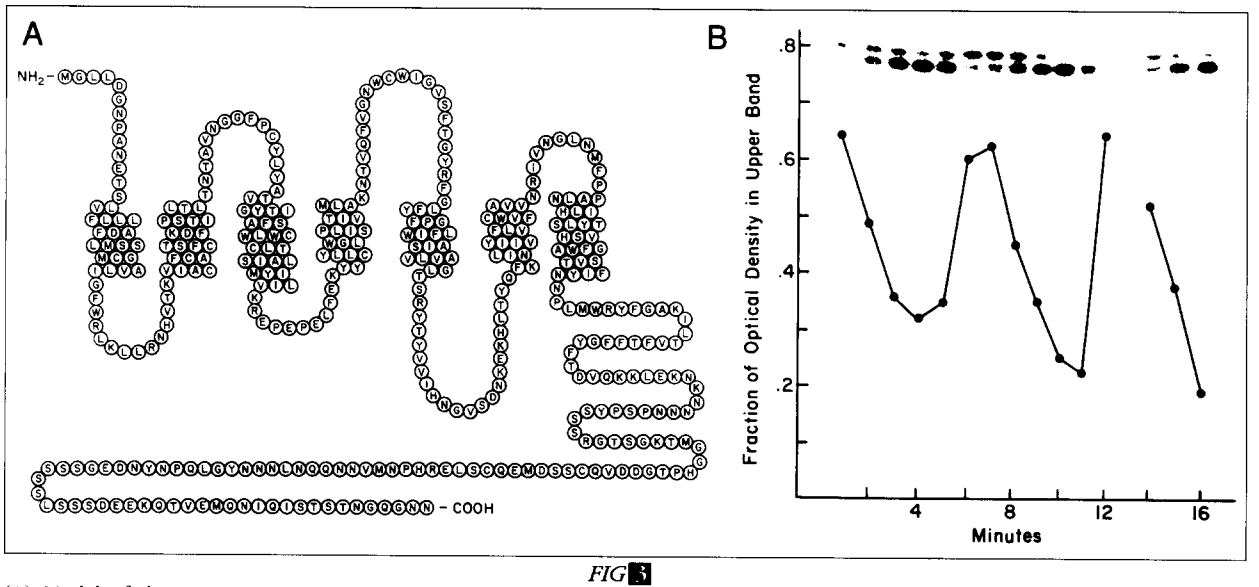

FIG 3

(A) Model of the cAMP receptor. The seven hydrophobic domains are arranged as α-helices in the lipid bilayer. The proposed extracellular domains are shown above the α-helices and the intracellular domains, including the serine-rich carboxy-terminus, are shown below these transmembrane helices. (B) Oscillatory behavior of the surface cAMP receptor. Cells were developed for 4 hours and oscillations in cAMP synthesis were monitored by measuring light scattering. Samples were taken at 1 min intervals during the oscillations, cells were photolabeled with ^{32}P-8-N_3-cAMP, analysed on SDS-PAGE, and autoradiographed. The autoradiographs were quantitated by scanning.

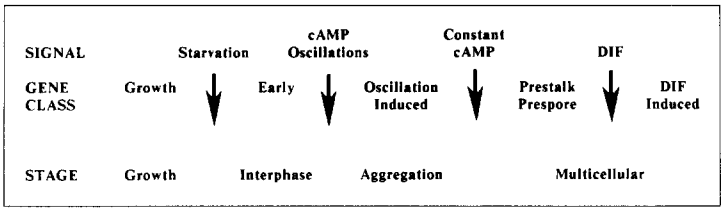

FIG 4

Time course and regulation of gene expression. Summary of RNA blot and nuclear run-on analysis of marker gene expression during development. The second line shows each of the classes of genes – growth, early, oscillation-induced, prestalk and prespore, and DIF-induced. The third line shows the developmental stage at which each class is expressed. The signals to which the gene classes respond, indicated by the arrows at the top, are produced by the cells during normal development. When cells are prevented from aggregating these cell–cell interactions do not occur and development is blocked. However, expression of all of the gene classes will occur if the signals are applied exogenously.

includes rhodopsin, the α- and β-adrenergic receptors, the serotonin receptor, and the muscarinic acetylcholine receptor.

On SDS-polyacrylamide gel electrophoresis, the cAMP receptor consists of a doublet of bands (molecular weights 40 000 and 43 000) that display a striking feature[19]: their relative intensity oscillates in synchrony with the spontaneous oscillations in cAMP signaling (Fig. 3B). When the oscillations are blocked, cells gradually attain a basal state and receptors are found predominantly as the smaller form; exogenous cAMP stimulates a rapid transition to the larger form[20]. This transition is accompanied by a dramatic increase in phosphorylation. Four to six serines, contained in the carboxy-terminal domain, are modified within a few minutes[21,22].

The adaptation properties of several rapid cellular responses such as myosin phosphorylation and adenylate cyclase activation have been carefully examined[23–27]. Several major properties of adaptation have emerged from these studies. Cells respond to increases in the fractional occupancy of receptors. When occupancy is held constant, responses subside within a few minutes. The magnitude of a response is proportional to the increment in occupancy. When the stimulus is withdrawn, adaptation decays with a half-life of several minutes and cells regain sensitivity in about 15 minutes. The pattern of modulation of phosphorylation is strongly correlated with the adaptation process[22,23]. As the level of phosphorylation approaches a new steady state, the physiological responses subside. When the stimulus is withdrawn, the cells regain sensitivity at the same rate as the decay in phosphorylation. Similar correlations of phosphorylation with desensitization are observed for rhodopsin and the β-adrenergic receptor. Thus, studies of adaptation mechanisms in this simple organism may apply directly to the G-protein-linked receptors in higher eukaryotic organisms.

Strong evidence that the surface cAMP receptor plays an essential role in morphogenesis and development has been obtained from observations of cell lines that fail to express the protein. These lines can be constructed by transformation with a vector that expresses antisense RNA (Ref. 18 and T. Sun *et al.*, in preparation). During starvation, the control cells aggregate and form streams in parallel with wild-type cells; the antisense transformants remain as a cell monolayer, without aggregating. The cAMP receptor protein is undetectable by immunoblot analysis. The cells display only very low levels of cAMP-binding sites (0.5%) and weak chemotactic and cAMP signaling responses. Few developmentally regulated genes are expressed. Thus the surface cAMP receptors are required for aggregation and for cells to progress through the developmental program.

Gene expression

Figure 4 illustrates a simplified classification of the genes expressed during development[5–9,28,29]. One class of genes is induced by starvation alone. Other classes require repeated application of cAMP for induction or are repressed by this treatment. Expression of the cell-type-specific genes, characteristic of prestalk or prespore cells, requires intermittent followed by continuous stimulation by cAMP. Finally, expression of a class of stalk genes requires DIF in addition to cAMP. There is an intriguing feature of the regulation by intermittent application of cAMP: low, constant concentrations of exogenous cAMP profoundly inhibit these genes and early development, while repeated application of cAMP at six-minute intervals, which mimics the endogenous oscillations, accelerates these processes. This suggests that the receptor-mediated responses that activate gene expression display adaptation and that the cAMP oscillator regulated by receptor-mediated phosphorylation (and dephosphorylation) may be an essential part of the developmental timer. In contrast, once the cells are made competent by repeated stimulation, the late cell-type-specific gene expression depends on continuous receptor occupancy. Thus, the responses that activate the late cell-type-specific genes must not adapt (see Fig. 4).

Morphogenesis and pattern formation

As the multicellular structure forms, a simple pattern appears along its axis. Cells within the posterior three-quarters of its length express 'prespore' markers and can be distinguished from those at the anterior, which do not express these markers. Since the anterior cells later differentiate into stalk cells, they are designated prestalk cells. The ratio of cell types can be altered pharmacologically and by mutation[30].

This pattern arises from both chemotactic cell sorting and through subsequent position-dependent differentiation. Cells that have undergone periods of starvation, that enter the developmental program from early stages of the cell cycle, or that have been grown on glucose-deficient medium preferentially sort to the anterior region. Subsequent prespore marker accumulation is confined to the posterior region of the slug, and, later, stalk cell differentiation takes place at the tip and ventral side of the structure, indicating that there are position-dependent developmental cues[4].

The tip of the slug displays properties of an embryonic 'organizer'. If a slug is severed just behind the tip,

the tissue connected to the tip continues to migrate. However, migration of the remainder ceases, the cell mass reforms a mound, and a new tip appears before morphogenesis continues. Excised tips, grafted to host slugs, define new axes and thereby cause several smaller slugs to form and separate. These organizational properties of the tip appear to derive from its capacity to generate oscillatory cAMP signaling[31,32].

The roles of adenosine and differentiation-inducing factor (DIF) in the multicellular structure are being explored. Adenosine appears to antagonize the actions of extracellular cAMP, and its production in anterior regions has been postulated to inhibit the formation of additional tips and to prevent anterior cells from expressing prespore genes[33]. DIF has been identified as 1-(3,5-dichloro-2,6-dihydroxy-4-methoxy-phenyl)-1-hexanone, a lipid-like compound that is essential for stalk cell differentiation. Although certain marker mRNAs have been shown to be dependent on DIF for expression and to be expressed only in anterior cells, DIF itself is found in higher levels in the posterior cells. Mutations that affect DIF production shift the proportion of cell types within the multicellular structure.

Conclusions

The mechanisms of signal transduction in eukaryotic microorganisms appear to be similar to those in mammalian cells, and thus these simple creatures provide convenient models for study of relevant signal transduction phenomena. The characterization of mutants such as *frigid A* and the *synags* and cAMP receptor antisense cell lines has defined a major signal transduction pathway. These studies have established that G-protein-linked signaling systems play a key role in the developmental program.

It has become clear that signal transduction through surface cAMP receptors participates in a multitude of developmental functions. Can a single surface receptor carry out so many functions? Our preliminary evidence suggests that there is a family of related surface cAMP receptor subtypes and that each subtype may be responsible for one or a few of the cAMP-mediated functions. We must suppose that expression of antisense RNA of the first subtype blocks expression of all the receptor subtypes, since the antisense cell lines are defective in most aspects of the developmental program.

Many of the events associated with signaling were first inferred from observations of cell monolayers which can form elegant ring and spiral patterns resembling the patterns produced in the Zabotinsky oscillatory chemical reaction (Fig. 1A). Embryologists should note, however, that these patterns are apparent only under favorable conditions. Frequently the aggregates resemble embryonic tissue in which the motions of the individual cells are difficult to trace. If similar signaling systems exist in higher animals, they may be difficult to detect. There are major apparent differences between *Dictyostelium* development and that of embryos. In *Dictyostelium*, a cell mass forms from the aggregation of a population of independent cells; in an embryo, a highly ordered cell mass forms from the division of a single cell. Whether the rules for the steps of morphogenesis and pattern formation that follow, and appear similar in each case, are more fundamental than these differences will determine the future usefulness of *Dictyostelium* as a model for development.

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